

Kyle Rood

Masters in Computer Science, Concentration in AI + Sustainability

kylerood16@gmail.com

<https://rooddood.github.io/website/>

<https://www.linkedin.com/in/kyle-rood/>

Work Experience and Skills

Freelance Video Production & Post-Production Editor

June 2026 - Now

Skills: Non-Linear Editing / Adobe Creative Cloud (Premiere Pro) / Audio Post-Production / Talent Direction

- Clipping and editing short and long for content for comedians, actors, and athletes
- Edit high-volume short-form and long-form video content tailored for digital distribution, broadcast, and promotional campaigns for talent, athletes, and performers.
- Manage end-to-end post-production workflows, including ingest, rough cuts, fine edits, audio mixing, color correction, and final multi-platform delivery specs.
- Collaborate directly with talent and creative directors on set and in post to ensure brand consistency, timing, and high production value.

Live A/V Producer and Front of House, *The DC Improv*

Apr. 2026 - Now

Skills: Public Speaking / Working w/ Talent / Live Production + Sound Reinforcement / Stage & Intelligent Lighting Control Board / FOH Operations / Stage Management / Live Event Execution

- Operate FOH audio mixing consoles and lighting control boards for high-profile, internationally touring talent at a premier national live venue.
- Manage live stage operations, signal routing, wireless microphone frequency allocation, and sound checks to guarantee optimal acoustics and zero downtime during performances.
- Deliver stage announcements, manage house dynamic audio levels, and coordinate seamlessly with touring crews, stage managers, and talent

Freelance Journalist and Technical Specialist, *The Washington Blade*

Apr. 2026 - Now

Skills: Writing / Wordpress / GCP + Ad Center / Social Media / Event Planning / Digital Content Strategy | WordPress Administration | GCP Deployment | Ad Tech & Analytics | Content Management

- Publish weekly investigative, interview, and feature stories focused on local Washington D.C. culture history
- Provide technical systems support for web content infrastructure, maintaining WordPress media delivery, digital asset workflows, and Google Cloud Platform deployment.
- Optimize digital ad operations and web performance tracking using Google Ad Manager and Analytics to drive viewer engagement and audience reach

Full Stack Implementation Engineer, *Strategic Solutions Research*

Dec. 2024- Now

Skills: Python / React / Next.JS / TypeScript / HuggingFace / BERT / AWS / LLMs (Anthropic Claude) / Wordpress /.NET / Media Data Architecture | Custom Visualization Tools | Automated Workflows | Python Pipeline Architecture | Next.js / React | GCP / AWS

- Architected and deployed dynamic data visualization and reporting tools in React, automating media client reporting processes and broadcast market analytics.
- Engineered custom survey and analytics software for radio stations and broadcast networks to streamline market research deployment and real-time data ingestion.
- Architected + developed advanced data visualization tools in React, automating client report generation
- Designed + built scalable Python ETL pipelines, reducing client reporting time + improving data accessibility
- Developed and deployed custom Topic Modeling + Sentiment Analysis models using BERT model via HuggingFace, increasing the depth of client insights and reducing operational costs

Full Stack AI Engineer, Hilltop-App

Feb. 2025- Mar. 2025; Feb. 2026 - April 2026

Skills: Python / Next.JS / TypeScript / Angular / GCP / LLMs (Gemini + OpenAI) / MCP / LangChain + RAG

- Rebuilt UI for a Next.js/Angular application in context-aware AI assistant, improving user experience + engagement
- Applied AI infrastructure (LangChain + RAG) to connect legislative data, automating contextual message generation
- Developed Python testing frameworks to ground agent responses, ensuring accuracy and context relevance
- Performed critical bug and system fixes, including resolving deployment issues and expired API keys.

Engineering Intern → Full Stack Engineer → Lead Engineer, Resultid

Sep. 2021- June 2024

Skills: Python / React / AWS / LLMs (OpenAI) / BERT / Docker / SQL / NoSQL / Git / CI/CD / Leadership

- Led + mentored growing team of engineers, establishing Agile development practices and maintaining high engineering standards using GitHub+AWS workflows (CI/CD)
- Designed + built a scalable data science application (from greenfield) leveraging AWS to process thousands of documents weekly
- Trained + hosted custom NLP algorithms (Sentiment Analysis, Topic Modeling, Semantic Search, etc.) on AWS EC2, increasing model performance + reducing latency
- Interfaced directly with customers to integrate feedback into features, resulting in an increase in user retention

Research Assistant → Lead ML Research Asst., The George Washington University

Dec. 2017 - May 2021

Skills: Research / Python / AWS / Computer Vision / Big Data / SQL / NoSQL (MongoDB)

- Collaborated on a DARPA project, developing a data-driven solution that reduced first responder time on-scene
- Full-Stack development of AMOS, a big data system that collected + categorized over 1 billion images from thousands of cameras
- Performed water depth estimation on a flooded city using multiband satellite imagery, providing critical data for disaster relief efforts
- Wrote, recorded, and edited auditory computer science exercises for introductory CS students

Python Teaching Assistant, The George Washington University

Sep. 2018 - May 2021

Skills: Python / Big Data / Databases / Data Science / Communication / Leadership / Teaching

- Designed + delivered curriculum on Python for data analytics including Numpy, Pandas, and Scikit-learn
- Empowered non-CS students to apply foundational programming concepts to real-world datasets
- Held regular office hours to provide one-on-one mentorship, increasing student confidence + comprehension

Front of House Employee, Maison Villatte (French Bakery)

Nov. 2025 - Feb 2026

- Managed all customer service and FOH operations, including drink preparation, customer relations, and maintenance

Education

The George Washington University

Masters in Computer Science w/ Concentration in Artificial Intelligence, **GPA: 4.0**

May 2021

Bachelors of Science in Computer Science, Minor in Sustainability, **GPA: 3.7**

May 2020

Skills Summary

Audio/Visual Tech: Adobe Audition, Premiere, Audacity, Cakewalk

Creativity: Film Photography, Creative writing (novel and script), Public Speaking (Standup, Model U.N.)

Languages + Technologies: Python, React, Next.JS, Angular, HTML/CSS, Java, Unity, Unreal Engine, C#, .NET

GenAI & Agents: Multi-Agent Orchestration (MCP/pydantic-AI), Prompt Engineering, RAG Systems, Model fine tuning

Large Language Models (LLMs): Gemini, Anthropic Claude, GPT, HuggingFace, LangChain

Core ML/AI: Natural Language Processing (NLP), Computer Vision, Sentiment + Topic Modeling, Semantic Search

AI Models: BERT, Whisper, DeepSeek

Data & Databases: ETL Pipelines, Vector (DocumentDB), SQL, NoSQL (MongoDB), Data ingestion/Scraping

Cloud & DevOps: Git/GitHub Workflows (CI/CD), AWS (Lambda, EC2, CodePipeline, etc.), GCP, Azure

DevOps: Docker, Kubernetes

Certifications: Sustainability Minor, SCRUM Master Certified, Apple Tech Support

Other technologies: Microsoft Office, Google Cloud

Solving Problems: Designing + implementing recipes (cooking), Making + selling clothes on Depop/Instagram/in person

Community: Volunteer with Don't Tell Comedy